

Pauli's Exclusion Principle :

No two electrons in an atom can have the same set of all the four quantum numbers, i.e. an orbital cannot have more than 2 electrons because three quantum numbers (principal, azimuthal and magnetic) may be same but the fourth must be different, i.e., spins must be in opposite directions.

Aufbau Principle :

Electrons are filled in various sub shells in order of their increasing energies. An orbital of lowest energy is filled first. The sequence of orbitals in order of their increasing energy is :

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d,

The energy of the orbitals is governed by $(n + l)$ rule

Hund's Rule :

No electron pairing takes place in the orbitals in a sub energy shell until each orbital is occupied by an electron with parallel spin.

Exactly half filled and fully filled orbital make the atoms more stable, i.e., p^3 , p^6 , d^5 , d^{10} , f^7 and f^{14} configurations are more stable.

LECTURE SHEET**EXERCISE-I**

(Introduction, fundamental particles, Atomic number, Mass number, Thomson's Atomic model, Rutherford's atomic model, Nature of light, EMR, Blackbody radiation, Photo electric effect, Bohr's theory, calculations of radius of orbit, velocity and energy of an electron)

LEVEL-I (MAIN)**Straight Objective Type Questions**

- Electron is a particle having a
 - negative charge of one unit and zero mass
 - positive charge of one unit and zero mass
 - negative charge of one unit and a mass of about 9.1×10^{-31} kg
 - negative charge of one unit and a mass of about 1.67×10^{-27} kg
- The value of e/m for an electron is
 - 1.78×10^8 c/g
 - 1.6724×10^{-24} c/g
 - 0.005486 c/g
 - 1.00866 c/g
- A and Z can be
 - negative
 - fractional
 - zero
 - whole number
- The number of protons, electrons and neutrons in $^{80}_{35}\text{Br}$ are respectively
 - 35, 35, 80
 - 35, 35, 45
 - 80, 80, 35
 - 45, 45, 35
- The nucleus of tritium consists of
 - 1 proton + 1 neutron
 - 1 proton + 3 neutrons
 - 1 proton + zero neutrons
 - 1 proton + 2 neutrons
- Sodium ion is isoelectronic with the atom
 - Mg^{2+}
 - Al^{3+}
 - Ne
 - N^{3-}

7. An atom differs from its ion in
 - 1) Nuclear charge
 - 2) Mass number
 - 3) Number of electrons
 - 4) Number of neutrons
8. In C^{14} isotope, the number of neutrons would be
 - 1) 6
 - 2) 14
 - 3) 8
 - 4) 10
9. The radius of the atom is of the order of
 - 1) 10^{-10} cm
 - 2) 10^{-13} cm
 - 3) 10^{-15} cm
 - 4) 10^{-8} cm
10. An oxide of nitrogen has a molecular weight of 30. Total number of electrons in one molecule of the compound is
 - 1) 15
 - 2) 30
 - 3) 45
 - 4) 60
11. When alpha particles are sent through a thin metal foil, most of them go straight through the foil because
 - 1) Alpha particles are much heavier than electrons
 - 2) Alpha particles are positively charged
 - 3) Most part of the atom is empty
 - 4) Alpha particles move with high velocity
12. Which of the following properties of a wave is independent of the other?
 - 1) Wave number
 - 2) Wave length
 - 3) Frequency
 - 4) Amplitude
13. Which of the following radiation following has highest wave number?
 - 1) Microwaves
 - 2) X - rays
 - 3) I.R. - rays
 - 4) Radiowaves
14. Which of the following relates a photon, both as a wave and as a stream of particles?
 - 1) $E = mc^2$
 - 2) Photoelectric effect
 - 3) Diffraction
 - 4) $E = h\nu$
15. Kinetic energy of photoelectrons is independent of the ____ of incident radiation.
 - 1) Wavelength
 - 2) Wave number
 - 3) Frequency
 - 4) Intensity
16. The energy of a photon is 3×10^{-12} ergs. What is its wavelength in nm ? ($h = 6.62 \times 10^{-27}$ erg. sec; $c = 3 \times 10^{10}$ cm.s $^{-1}$)
 - 1) 662
 - 2) 1324
 - 3) 66.2
 - 4) 6.62
17. The frequency associated with photon of radiation having a wavelength of $6000\text{\AA}$ is
 - 1) 5×10^{14} Hz
 - 2) 5×10^{10} Hz
 - 3) 2×10^{14} Hz
 - 4) 5×10^{15} Hz
18. Bohr's theory is applicable to
 - 1) Li^{+2}
 - 2) Li^+
 - 3) He^+
 - 4) Both 1 and 3
19. Bohr's theory is not applicable to
 - 1) H
 - 2) He^+
 - 3) Be^{3+}
 - 4) H^+
20. (A) : Bohr's orbits are called stationary orbits
 (R) : Electrons remain stationary in these orbits
 - 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 - 3) (A) is true but (R) is false
 - 4) (A) is false but (R) is true

21. Energy of an electron in n^{th} Bohr orbit is given as

- 1) $-\frac{n^2 h^2}{4\pi^2 m Z e^2}$ 2) $-\frac{2\pi^2 Z^2 m e^4}{n^2 h^2}$ 3) $-\frac{2\pi Z e^2}{n h}$ 4) $-\frac{n^2 h^2}{2\pi^2 Z^2 m e^4}$

22. The energy of the electron when it is at an infinite distance from the nucleus is

- 1) Infinity 2) Zero
3) Minimum 4) Can not be predicted

23. According to Bohr's theory, when ever the electron drops from a higher energy level to a lower energy level, the frequency of radiation emitted is related to the energy change as

- 1) $\lambda = \frac{h}{m v}$ 2) $m v r = \frac{n h}{2\pi}$ 3) $v = \frac{\Delta E}{h}$ 4) $v = \frac{h}{\Delta E}$

24. Radius of 3rd Bohr orbit of hydrogen atom

- 1) 6.529 Å° 2) 2.116 Å° 3) 4.761 Å° 4) 8.464 Å°

25. Velocity of the electron in the 1st Bohr orbit of H-atom is

- 1) 2.18×10^8 cm/sec 2) 2.18×10^8 m/sec
3) 2.18×10^{16} cm/sec 4) 36559×10^8 cm/sec

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. An α - particle having kinetic energy 5 MeV falls on a Cu-foil. The shortest distance from the nucleus of Cu to which α - particle reaches is (Atomic no. of Cu = 29, $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

- a) $2.35 \times 10^{-13} \text{ m}$ b) $1.67 \times 10^{-14} \text{ m}$ c) $5.98 \times 10^{-15} \text{ m}$ d) $1.67 \times 10^{-16} \text{ m}$

2. If the energy of H-atom in the ground state is $-E$, the velocity of photo-electron emitted when a photon having energy E_p strikes a stationary Li^{2+} ion in ground state, is given by:

- a) $v = \sqrt{\frac{2(E_p - E)}{m}}$ b) $v = \sqrt{\frac{2(E_p + 9E)}{m}}$ c) $v = \sqrt{\frac{2(E_p - 9E)}{m}}$ d) $v = \sqrt{\frac{2(E_p - 3E)}{m}}$

3. The energy of a I, II and III energy levels of a certain atom are E , $\frac{4E}{3}$ and $2E$ respectively A photon of wavelength λ is emitted during a transition from III to I. What will be the wavelength of emission for transition II to I?

- a) $\frac{\lambda}{2}$ b) λ c) 2λ d) 3λ

4. The increasing order of e/m values for electron, proton, neutron and alpha particle is

- a) e, p, n, α b) n, p, e, α c) n, p, α , e d) n, α , p, e

5. The constancy of e/m ratio for electron shows that

- a) Electron's mass is $1/1837^{\text{th}}$ of the mass of proton
b) Electrons are universal particles of all matter
c) Electrons are produced in discharge tubes only
d) None of the above

6. The e/m of proton is

- a) $1.78 \times 10^8 \text{ c/g}$ b) $9.57 \times 10^4 \text{ c/g}$ c) $19.14 \times 10^4 \text{ c/g}$ d) $0.478 \times 10^4 \text{ c/g}$

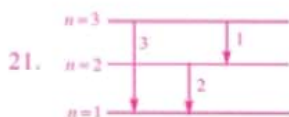
7. Number of protons in the nucleus of carbon atom is
a) 7 b) 8 c) 4 d) 6
8. The number of nucleons present in chlorine-37 is
a) 17 b) 20 c) 54 d) 37
9. The number of neutrons in the dipositive zinc ion (Mass number of Zn = 65)
a) 35 b) 33 c) 65 d) 67
10. The charge of an electron is 1.6×10^{-19} coulombs. What will be the value of charge on Na^+ ion
a) 1.6×10^{-19} C b) 3.2×10^{-19} C c) 2.4×10^{-19} C d) $11 \times 1.6 \times 10^{-19}$ C
11. Maximum sum of the number of neutrons and protons in an isotope of hydrogen
a) 6 b) 5 c) 4 d) 3
12. The incorrect statement regarding cathode rays is
a) They travel in straight line
b) They depend on the nature of the gas
c) They are deflected by magnetic as well as electric fields
d) They produce mechanical effects
13. According to Planck's Quantum theory, the correct statements are
I) The vibrating particle in the black body does not emit continuously
II) Radiation is emitted in the form of small packets called Quanta
III) Energy associated with emitted radiations is inversely proportional to frequency.
IV) The emitted radiant energy is propagated in the form of waves.
a) I, II, III b) II, III c) I, II, IV d) II, IV, III
14. Which of the following statements is incorrect ?
a) Particle nature of radiations can be experimentally demonstrated by Compton effect
b) Wave nature of electrons can be experimentally demonstrated by diffraction experiment
c) The value of Planck's constant, h is 6.62×10^{-34} J.s
d) Intensity of light is directly proportional to its frequency.
15. As an electron is brought from an infinite distance close to nucleus of atom, the energy of electron
a) Increases to a greater +ve value b) Decreases to a smaller +ve value
c) Increases to a smaller -ve value d) Decreases to a greater -ve value
16. Rutherford's alpha ray scattering experiment showed for the first time that the atom has
a) Nucleus b) Proton c) Electron d) Neutron
17. In electromagnetic radiation, which of the following has greater wavelength than visible light?
a) U.V-rays b) I.R-rays c) Gamma rays d) X-rays
18. Which of the following is not an electromagnetic radiation?
a) Gamma rays b) Alpha rays c) Radio waves d) X-rays

More than One correct answer Type Questions

19. In Rutherford's gold leaf experiment, the scattering of α -particles takes place. In this process :
a) coulombic force is involved b) nuclear force is involved
c) path of α -particle is parabolic d) path of α -particle is hyperbolic

20. According to Bohr's model of atom

- a) The radius of n^{th} orbit $\propto n^2$
- b) The total energy of electron in n^{th} orbit $\propto \frac{1}{n}$
- c) The angular momentum electron is integral multiples of $h/2\pi$
- d) The magnitude of potential energy of an electron in an orbit greater than kinetic energy



21. For above transitions in hydrogen like atoms, select the incorrect relation(s).

- a) $v_3 = v_1 + v_2$
- b) $v_3 = \frac{v_1 v_2}{v_1 + v_2}$
- c) $\lambda_3 = \lambda_1 + \lambda_2$
- d) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

22. If the electron of the hydrogen atom is replaced by another particle of same charge but of the double mass, then :

- a) radii of different shells will increase
- b) energy gap between two levels will become double
- c) ionization energy of the atom will be double
- d) speed of new particle in a shell will be lesser than the speed of electron in the same shell

23. Let r , u and E are the radius of the orbit, speed of the electron and the total energy of the electron respectively. Which of the following quantities are proportional to the quantum number ' n ' ?

- a) rE
- b) ur
- c) $\frac{u}{E}$
- d) $\frac{r}{E}$

24. Select the correct statement(s):

- a) An electron near the nucleus is attracted by the nucleus and has a low potential energy
- b) According to Bohr's theory, an electron continuously radiate energy if it stayed in out orbit
- c) Bohr's model could not explain the spectra of multielectron atoms
- d) Bohr's model was the first atomic model based on quantisation of energy

Linked Comprehension Type Questions

Passage-I :

A metal is irradiated with a radiation of wavelength 300 nm. Then photo electrons found to have velocity of $3.7 \times 10^5 \text{ m/s}$. When a graph is drawn between K.E of photo electrons and frequency of radiation a straight line is obtained cutting X-axis.

25. What is the K.E. of photo electrons

- a) $6.23 \times 10^{-20} \text{ J}$
- b) $6.25 \times 10^{-22} \text{ J}$
- c) $6.625 \times 10^{-18} \text{ J}$
- d) $6.625 \times 10^{-19} \text{ J}$

26. The value of X-intercept

- a) $9 \times 10^4 \text{ Hz}$
- b) $10.2 \times 10^5 \text{ Hz}$
- c) $8.6 \times 10^5 \text{ Hz}$
- d) $9 \times 10^{14} \text{ Hz}$

27. Difference between incident frequency and threshold frequency is

- a) $9.4 \times 10^{13} \text{ Hz}$
- b) $8.5 \times 10^{12} \text{ Hz}$
- c) $9.4 \times 10^{12} \text{ Hz}$
- d) $8.4 \times 10^{10} \text{ Hz}$

Passage-II :

To explain the nature of radiations, James Clark Maxwell in 1864 put forward 'Electromagnetic wave theory'. This theory could explain the phenomena of interference and diffraction but could not explain phenomena of black body radiation and photoelectric effect. Also, after the advent of electromagnetic wave theory, Rutherford's model of atom suffered a serious drawback. To explain the phenomena of black body radiation and photoelectric effect, Max Planck in 1900 put forward Planck's quantum theory. Based on this theory, Neils Bohr in 1913 put forward Bohr's model of atom which could overcome the drawback of Rutherford's model and also explain the line spectra of elements, especially the line spectra of hydrogen and hydrogen-like particles.

28. Which of the following statements is incorrect about electromagnetic wave theory ?
 a) Energy is emitted continuously from the source
 b) Radiations are associated with electric and magnetic fields
 c) Radiations can travel even through vacuum
 d) The velocity of any radiation depends upon the nature of the radiation.
29. Kinetic energy of the electron ejected when yellow light of frequency $5.2 \times 10^{14} \text{ s}^{-1}$ falls on the surface of potassium metal (threshold frequency = $5 \times 10^{14} \text{ s}^{-1}$) is
 a) $1.325 \times 10^{-13} \text{ J}$ b) $1.325 \times 10^{-20} \text{ J}$ c) $1.325 \times 10^{-27} \text{ J}$ d) $1.325 \times 10^{-34} \text{ J}$

Matrix Matching Type Questions**30. Column-I**

- A) Isotopes
 B) Isobars
 C) Isotones
 D) Isosters

Column-II

- P) Same no. of atoms and same no. of electrons
 Q) Different no. of protons but same no. of neutrons
 R) Different no. of protons and different no. of neutrons
 S) Same no. of protons but different no. of neutrons

31. Column-I

- A) $\phi < \phi_0$
 B) $\lambda < \lambda_0$
 C) $\phi > \phi_0$
 D) $\lambda > \lambda_0$

Column-II

- P) photoelectron emitted
 Q) no photocurrent
 R) K.E of photoelectron = $h(\phi - \phi_0)$
 S) velocity of photo electron = $\sqrt{\frac{2ch}{m} \left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right)}$

32. Column-I

- A) Radius of n^{th} orbit
 B) Energy of electron in n^{th} orbit
 C) Velocity of electron in n^{th} orbit
 D) Angular momentum of electron

Column-II

- P) inversely proportional to Z
 Q) proportional to n^2
 R) inversely proportional to n
 S) Proportional to n

33. According to Bohr's theory, E_n = total energy, K_n = kinetic energy, V_n = potential energy, r_n = radius of n^{th} orbit. Match the following.

Column-I

- A) $V_n/K_n = ?$
 B) $r_n \propto E_n^x, x = ?$
 C) Angular momentum in lowest orbital
 D) $\frac{1}{r_n} \propto Z^y, y = ?$

Column-II

- P) 0
 Q) -1
 R) -2
 S) 1

Integer Type Questions

34. Find the ratio of energy of a photon of $2000\text{\AA}$ wavelength radiation to that of $4000\text{\AA}$ radiation.
 35. A light source of wavelength λ illuminates a metal and ejects photo-electrons with $(K.E)_{\text{max}} = 1\text{eV}$. Another light source of wave length $\frac{\lambda}{3}$, ejects photo-electrons from same metal with $(K.E)_{\text{max}} = 4\text{eV}$, the value of work function is $0.1x$. Then the value of x is ____
 36. Assume that $2 \times 10^{-17}\text{ J}$ of light energy is needed by the interior of the human eye to see an object. The number of photons of yellow light with $\lambda = 595.2\text{nm}$ are needed to generate this minimum energy is $12x$. Then the value of x is ____.
 37. Number of photons of light with a wave length of 4000 pm necessary to provide one joule of energy approximately $x \times 10^{16}$. Then x is ____
 38. The velocity of an electron in a certain Bohr's orbit of H-atom bears the ratio 1 : 275 to the velocity of light. Then find the quantum number (n) of the orbit ?
 39. When an electron is transited from $2E$ to E energy level, the wavelength of resultant photon produced is λ_1 . If electronic transition involves $\frac{4}{3}E$ to E level, the wavelength of resultant photon is λ_2 . The correct relation between λ_1 & λ_2 is $\lambda_2 = x\lambda_1$. Then find the value of x ?
 40. Given $r_{n+1} - r_{n-1} = 2r_n$, where r_n, r_{n-1}, r_{n+1} are Bohr radius for hydrogen atom in $n^{\text{th}}, (n-1)^{\text{th}}$ and $(n+1)^{\text{th}}$ shell respectively. Calculate the value of n .

EXERCISE-II

(H- spectrum, Rydberg equation, de broglies concept of matter waves, Heisenberg uncertainty principle, schrodinger wave equation, orbitals)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. The equation corresponding to the wave number of spectral lines in Pfund series is
 1) $R\left[\frac{1}{4^2} - \frac{1}{5^2}\right]$ 2) $R\left[\frac{1}{3^2} - \frac{1}{4^2}\right]$ 3) $R\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$ 4) $R\left[\frac{1}{5^2} - \frac{1}{6^2}\right]$
 2. The value of Rydberg constant is
 1) 109677 cm^{-1} 2) $109700\text{ cm}^{-1}\text{ s}^{-1}$
 3) 10968 cm^{-1} 4) 10970 m

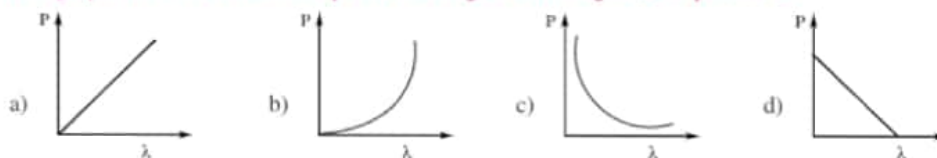
3. A spectral line of hydrogen with $\lambda = 4938\text{\AA}$ belongs to the series
 1) Lyman 2) Balmer 3) Paschen 4) Pfund
4. Among the first lines of Lyman, Balmer, Paschen and Brackett series in hydrogen atomic spectra, which has higher energy?
 1) Lyman 2) Balmer 3) Paschen 4) Brackett
5. What are the values of n_1 and n_2 respectively for H_β line in the Lyman series of hydrogen atomic spectrum?
 1) 3 and 5 2) 2 and 3 3) 1 and 3 4) 2 and 4
6. The fourth line of the Balmer series corresponds to the electronic transition between two orbits of the H atom. Identify the orbits.
 1) 3 and 1 2) 5 and 1 3) 5 and 2 4) 6 and 2
7. Total number of spectral lines when electron jumps from 8th orbit to 2nd orbit
 1) 6 2) 36 3) 21 4) 38
8. Supposing the electron is present in the 4th energy level of H- atom. When the electron returns to ground state the possible transitions would be :
 1) $4 \rightarrow 1$ 2) $4 \rightarrow 2, 2 \rightarrow 1$ 3) $4 \rightarrow 3, 3 \rightarrow 2, 2 \rightarrow 1$ 4) All the above
9. The de-Broglie's equation treats an electron to be
 1) a particle 2) a wave 3) ray 4) both (1) and (2)
10. Wavelength of the wave associated with a moving electron
 1) Decreases with increase in speed of electron
 2) Increases with increase in speed of electron
 3) Remains same irrespective of speed of electron 4) is zero
11. The uncertainty principle and the concept of wave nature of matter were proposed by _____ and _____ respectively
 1) Pauli, Hund 2) Heisenberg, Aufbau
 3) Heisenberg, de Broglie 4) Heisenberg, Planck
12. Bohr's postulate that $mvr = \frac{nh}{2\pi}$ is proved mathematically by
 1) Pauli's exclusion principle 2) de Broglie wave nature of the electron
 3) Heisenberg's uncertainty principle 4) Sommerfeld theory
13. The momentum of a particle of wave length $1\text{\AA}$ is
 1) $6.625 \times 10^{-27} \text{ g. cm.s}^{-1}$ 2) $6.625 \times 10^{-19} \text{ g. cm.s}^{-1}$
 3) $6.625 \times 10^{-16} \text{ g. cm.s}^{-1}$ 4) $6.625 \times 10^{-23} \text{ g. cm.s}^{-1}$
14. If uncertainty in position and velocity are equal then uncertainty in momentum will be:
 1) $\frac{1}{2}\sqrt{\frac{mh}{\pi}}$ 2) $\frac{1}{2}\sqrt{\frac{h}{\pi m}}$ 3) $\frac{h}{4\pi m}$ 4) $\frac{mh}{4\pi}$
15. Be^{+3} and a proton are accelerated by the same potential, their de-Broglie wavelengths have the ratio (assume mass of proton = mass of neutron)
 1) 1:2 2) 1:4 3) 1:1 4) 2:1

LEVEL-II (ADVANCED)

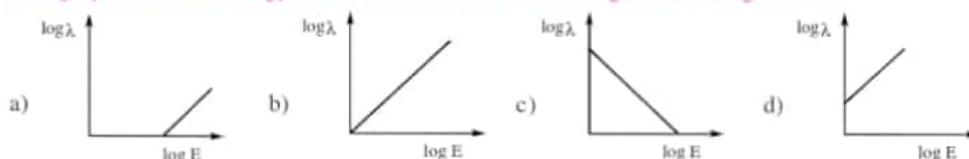
Straight Objective Type Questions

- Which of the following statements regarding spectral series is correct?
 - The lines in the Balmer series correspond to the electronic transition from higher energy level to $n = 1$ energy level.
 - Paschen series appears in the infra-red-region
 - The lines of Lyman series appear in the visible region
 - Transition from higher energy levels to 4th energy level produces Pfund series which fall in the infrared region.
- In a series in the line spectrum of hydrogen, the wavelength of radiation is $6.563\text{\AA}$. The name of the series and the orbits in which electron transition takes place are
 - Balmer series, 3rd to 2nd orbit
 - Lyman series, 2nd to 1st orbit
 - Pfund series, 6th to 5th orbit
 - Paschen series, 4th to 3rd orbit
- The frequency of the spectral line obtained when the electron in $n = 3$ of Hydrogen atom drops to the ground state is
 - 2.925×10^{15} Hertz
 - 2.925×10^{13} Hertz
 - 2.925×10^{14} Hertz
 - 36559×10^{10} Hertz
- (A) : The energy of ultraviolet radiation is greater than the energy of infrared radiation
 (R) : The velocity of ultraviolet radiation is greater than the velocity of infrared radiation
 - Both A and R are true and R is the correct explanation of A
 - Both A and R are true but R is not the correct explanation of A
 - A is true and R is false
 - R is true and A is false
- Which of the following transitions will have minimum wavelength ?
 - $n_4 \rightarrow n_1$
 - $n_2 \rightarrow n_1$
 - $n_4 \rightarrow n_2$
 - $n_3 \rightarrow n_1$
- A proton and an electron are accelerated by the same potential difference. If λ_e and λ_p denote the De-broglie wavelength of electron and proton then
 - $\lambda_e = \lambda_p$
 - $\lambda_e < \lambda_p$
 - $\lambda_e > \lambda_p$
 - No relation between λ_e and λ_p
- Velocity of helium atom at 300K is 2.40×10^2 meter per second. What is its wavelength? (He = 4)
 - 0.416 nm
 - 0.83 nm
 - 803 Å
 - 8000 Å

- The graph between momentum p and de-Broglie wavelength λ of photon is



- The graph between energy of an electron and its de-Broglie wavelength λ is



10. A particle X moving with a certain velocity has a de Broglie wave length of $1\text{ \AA}$. If particle Y has a mass of 25% that of X and velocity 75% that of X, de Broglie's wave length of Y will be
 a) $3\text{ \AA}$ b) $5.33\text{ \AA}$ c) $6.88\text{ \AA}$ d) $48\text{ \AA}$
11. Which of the following statement is incorrect ?
 a) The third quantum shell can hold a maximum of 18 electrons
 b) An electron falling to the same energy level from any higher level always emits the same quantum of energy
 c) The Balmer series of lines is in the visible region of the emission spectrum of hydrogen atom
 d) The electron of hydrogen atom in its ground state remains in the first quantum shell
12. The Schrodinger wave equation for hydrogen atom is $\Psi_{2s} = \frac{1}{4\sqrt{2}\pi} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r_0}{a_0}\right) e^{-r/a_0}$ where a_0 is Bohr's radius. If the radial node in $2s$ be at r_0 would be equal to:
 a) $\frac{a_0}{2}$ b) $2a_0$ c) $\sqrt{2}a_0$ d) $\frac{a_0}{\sqrt{2}}$
13. The Schrodinger wave equation for hydrogen atom is

$$\Psi(\text{radial}) = \frac{1}{16\sqrt{4}} \left(\frac{z}{a_0}\right)^{3/2} [(\sigma - 1)(\sigma^2 - 8\sigma + 12)] e^{-\sigma/2}$$
 where a_0 and z are the constant in which answer can be expressed and $\sigma = \frac{2zr}{a_0}$ minimum and maximum position of radial nodes from nucleus are
 a) $\frac{a_0}{Z}, \frac{3a_0}{Z}$ b) $\frac{a_0}{2Z}, \frac{a_0}{Z}$ c) $\frac{a_0}{2Z}, \frac{3a_0}{Z}$ d) $\frac{a_0}{2Z}, \frac{4a_0}{Z}$
14. For a $3s$ - orbital $\Psi(3s) = \frac{1}{9\sqrt{3}} \left(\frac{1}{a_0}\right)^{3/2} (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$; where $\sigma = \frac{2rZ}{3a_0}$. What is the maximum radial distance of node from nucleus?
 a) $\frac{(3 + \sqrt{3})a_0}{z}$ b) $\frac{a_0}{z}$ c) $\frac{3(3 + \sqrt{3})a_0}{2z}$ d) $\frac{2a_0}{z}$

More than One correct answer Type Questions

15. Which of the following statements is/are correct ?
 a) The absorption spectrum is formed due to absorbing radiant energy by the matter in lower energy states
 b) The emission spectrum is formed due to the emission of radiant energy by the excited matter
 c) Hydrogen spectrum is an example of line spectrum
 d) Li^+ ion spectrum is equivalent to H atom spectrum
16. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$ where n_1 and n_2 are the principal quantum numbers of the two states. Assume the Bohr model to be valid, if the time period of the electron in the initial state is eight times that in the final state, then the possible values of n_1 and n_2 are
 a) $n_1 = 4, n_2 = 2$ b) $n_1 = 8, n_2 = 2$
 c) $n_1 = 8, n_2 = 1$ d) $n_1 = 6, n_2 = 3$

17. Whenever a hydrogen atom emits a quantum in the visible region of Balmer series :
- it may emit another quantum in Balmer series again
 - it must emit another quantum in ultraviolet region of Lyman series
 - second quantum of wavelength 122nm will be emitted
 - second quantum will be emitted in Paschen series i.e., near infrared region
18. Which of the following species are expected to give the line spectrum similar to that of hydrogen atom?
- He(g)
 - Li²⁺(g)
 - He⁺(g)
 - D(g)
19. Hydrogen atom has
- half filled subshell
 - half filled shell
 - one electron in valence shell
 - half filled orbital
20. Select the correct statement(s)
- The phenomena of diffraction of light can only be explained by assuming that light behaves as waves
 - de-Broglie postulate the dual character existed with matter
 - In his atomic model Bohr considered electron as a particle
 - Wave nature of electrons was obtained when diffraction rings were observed protographically when a stream of protons was passed through a metal foil
21. False statement about the de-Broglie's wavelength of an electron in the first Bohr's orbit is
- equal to the circumference of the first orbit
 - equal to twice the circumference of the first orbit
 - equal to half the circumference of the first orbit
 - equal to one fourth of the circumference of the first orbit
22. Which is correct in case of p-orbitals ?
- They are spherically symmetrical
 - They have strong directional character
 - They are three fold degenerate orbitals
 - Their charge density along x, y and z-axes are zero.

Linked Comprehension Type Questions

Passage:

de-Broglie proposed the dual nature of electron and put forward his wave concept.

The wave length of electron in an orbit was given by $\lambda = \frac{h}{mv}$

23. The de-Broglie's wavelength (λ) of the electron subjected to an accelerating potential of V volts is given by
- $\frac{eh}{\sqrt{2mV}}$
 - $\frac{h}{\sqrt{2meV}}$
 - $\frac{h}{\sqrt{2me}} \times V$
 - $\frac{mh}{\sqrt{2eV}}$

24. The frequency of the matter wave of a particle is given by

- a) $\frac{K.E}{h}$ b) $\frac{2K.E}{h}$ c) $\frac{K.E}{2h}$ d) $\frac{h}{2K.E}$

25. The circumference of 3rd Bohr orbit in H-atom is

- a) 3×10^{-7} cm b) 3×10^{-8} cm
c) 3×10^{-6} cm d) 4.3×10^{-9} cm

26. The wavelength of moving electron in 3rd Bohr orbit of H-atom is

- a) 1×10^{-9} m b) 2×10^{-7} m
c) 1×10^{-7} m d) 1×10^{-8} m

Matrix Matching Type Questions

27. In case of hydrogen spectrum wave number is given by $\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ where $n_1 < n_2$

Column-I

- A) Lyman series
B) Balmer series
C) Pfund series
D) Brackett series

Column-II

- P) $n_2 = 2$
Q) $n_2 = 3$
R) $n_2 = 6$
S) $n_2 = 5$

28. If the shortest wavelength of spectral line of H-atom in Lyman series is x, Then match the following for Li^{2+}

Column-I

- A) Shortest wavelength in Lyman series
B) Longest wavelength in Lyman series
C) Shortest wavelength in Balmer series
D) Longest Wavelength in Balmer series

Column-II

- P) $\frac{4x}{5}$
Q) $\frac{4x}{9}$
R) $\frac{x}{9}$
S) $\frac{4x}{27}$

Integer Type Questions

29. The wavelength of a certain line in Balmer Series is observed to be $4329 \text{ \AA}$. To what value of 'n' does this correspond ? ($R_H = 109678 \text{ cm}^{-1}$) ($Z = 1$)
30. The ratio of wave number of the limiting line of Lyman series to the limiting line of Balmer series in Hydrogen spectrum is ____
31. Calculate the number of waves made by a Bohr's electron in one complete revolution in its 3rd orbit of H-atom.

EXERCISE-III

(Quantum numbers, electron probability graphs, Hund's rule, Aufbau principle, Pauli's principle, electronic configurations of atoms and ions, abnormal configurations of Cr, Mn, magnetic behavior)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The azimuthal quantum number indicates ____ of the orbital
1) Size 2) Shape 3) Orientation 4) Spin
- Which of the following is indicated by the magnetic quantum number?
1) Size 2) Shape 3) Spatial orientation 4) Spin
- The spin quantum number has a value of
1) $1/2$ 2) $+1/2$ 3) $-1/2$ 4) either $+1/2$ or $-1/2$
- Which one of the following expressions represent the electron probability function (D)
1) $4\pi r dr \psi^2$ 2) $4\pi r^2 dr \psi$ 3) $4\pi r^2 dr \psi^2$ 4) $4\pi r dr \psi$
- Radial part of the wave function depends on quantum numbers
1) n and s 2) l and m 3) l and s 4) n and l
- Which of the following is correct with respect to 'p' orbitals?
1) Spherical 2) Strong directional character
3) Five fold degenerate 4) No directional character
- The maximum number of electrons accommodated in 5f orbitals
1) 5 2) 10 3) 14 4) 18
- The maximum probability of finding an electron of a particular energy in an orbital is about
1) 80% 2) 85% 3) 95% 4) 99%
- Number of nodal spaces in 4s orbital is
1) 0 2) 1 3) 3 4) 4
- (A) : The p-orbital has dumb-bell shape
(R) : Electron present in p-orbital can have any one of the three values of magnetic quantum numbers (0, +1, -1)
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false 4) (A) is false but (R) is true
- No two electrons in an orbital can have parallel spin. This statement emerges from
1) Hund's rule 2) Aufbau principle
3) Pauli's exclusion principle 4) $(n + l)$ rule
- Electrons never pair, if there are empty orbitals in a given sub-shell. This is
1) Aufbau principle 2) Pauli's exclusion principle
3) Hund's rule of maximum multiplicity 4) Heisenberg's uncertainty principle

13. Which of the following explains the sequence of filling electrons in different subshells?
 1) Hund's rule 2) Aufbau principle 3) Pauli's principle 4) All of these
14. Nitrogen atom has 3 unpaired electrons in its ground state. It can be explained by
 1) Auf bau principle 2) Pauli's principle 3) Hund's rule 4) All of these
15. The electronic configuration of sodium is
 1) [Ne]3s² 2) [Ne]3s¹ 3) [Ar]4s¹ 4) [Ar]4s²
16. Which of the following configuration may represent the ground state of nitrogen atom?
 1) $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ 2) $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ 3) $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ 4) $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$
17. Electronic configuration of the element with atomic number 56 and mass number 138 is
 1) [Xe]6s² 2) [Kr]5s² 3) [Xe]6s² 6p² 4) [Xe]3d² 5d²
18. In potassium the order of energy levels is
 1) 4s > 3d 2) 4s < 3d 3) 4s < 3p 4) 4s = 3d
19. The number of d-electrons retained in Fe²⁺ (At. no. of Fe = 26) ion is
 1) 3 2) 4 3) 5 4) 6
20. Which of the following is the correct electronic configuration of Fe³⁺ ion ? (Z for Fe = 26)
 1) 1s², 2s², 2p⁶, 3s², 3p⁶, 3d⁴ 4s¹ 2) 1s², 2s², 2p⁶, 3s², 3p⁶, 3d³ 4s²
 3) 1s², 2s², 2p⁶, 3s², 3p⁶, 3d⁵ 4) 1s², 2s², 2p⁶, 3s², 3p⁶, 3d⁵, 4s², 4p³
21. Number of unpaired electrons in the electrmonic configuration 1s²2s²2p⁴ is
 1) 2 2) 3 3) 4 4) 6

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. In a multi - electron atom, which of the following orbitals described by the three quantum numbers will have the same energy in the absence of magnetic and electric field
 i) n = 1, l = 0, m = 0 ii) n = 2, l = 0, m = 0 iii) n = 2, l = 1, m = 1
 iv) n = 3, l = 2, m = 1 v) n = 3, l = 2, m = 0
 a) i and ii b) ii and iii c) iii and iv d) iv and v
2. An electron has magnetic quantum number as '-3'. Its principal quantum number is
 a) 3 b) 2 c) 1 d) 4
3. Match the following.
- | | |
|----------------------------|--|
| LIST-I | LIST-II |
| a) n = 2, l = 1, m = -1 | p) 2p _x or 2p _y |
| b) n = 4, l = 2, m = 0 | q) 4dz ² |
| c) n = 3, l = 1, m = ±1 | r) 3p _x or 3p _y |
| d) n = 4, l = 0, m = 0 | s) 4s |
| e) n = 3, l = 2, m = ±2 | t) 3dx ² - y ² or 3dxy |
| a) a-q, b-r, c-p, d-s, e-t | b) a-t, b-r, c-s, d-p, e-t |
| | c) a-p, b-q, c-r, d-s, e-t |
| | d) a-s, b-t, c-r, d-s, e-p |

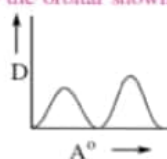
4. The quantum numbers of four electrons (e_1 to e_4) are given below

	n	l	m	s		n	l	m	s
e_1	3	0	0	+1/2	e_2	4	0	1	1/2
e_3	3	2	2	-1/2	e_4	3	1	-1	1/2

The correct order of decreasing energy of these electrons is:

- a) $e_4 > e_3 > e_2 > e_1$ b) $e_2 > e_3 > e_4 > e_1$ c) $e_3 > e_2 > e_4 > e_1$ d) $e_1 > e_3 > e_4 > e_2$
5. Which orbital gives an electron the greatest probability of closer to the nucleus ?
 a) 3p b) 3d c) 3s d) equal
6. Which one of the following statements is correct ?
 a) 2S orbital is spherical with two nodal planes
 b) The de-Borglie wavelength of a particle of mass 'm' and velocity 'v' is equal to mv/h
 c) The principal quantum number (n) indicates the shape of the orbital
 d) The electronic configuration of phosphorus is given by $[\text{Ne}] 3s^2 3p_x^1 3p_y^1 p_z^1$
7. The set of quantum numbers 'n' and 'l' possible for the orbital shown in the radial probability curve are

- a) $n = 3; l = 2$
 b) $n = 4; l = 1$
 c) $n = 2; l = 0$
 d) $n = 3; l = 3$



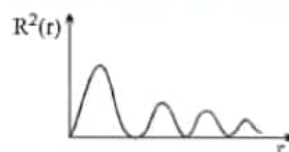
8. From the following observations predict the type of orbital :

Observation 1: x y plane acts as a nodal plane

Observation 2: The angular function of the orbital intersect the three axis at origin only.

Observation 3: $R^2(r)$ vs r curve is obtained for the orbital is

- a) $5p_z$
 b) $6d_{xy}$
 c) $6 dx^2 - y^2$
 d) $6 d_{yz}$



More than One correct answer Type Questions

9. Choose the correct statement.
 a) Splitting of spectral lines in magnetic field is due to presence of degenerate orbitals
 b) In the presence of electricfield, energy value of P_x , P_y & P_z of same orbit are different
 c) Degenerate orbitals differs only in their orientation
 d) degenerate orbitals have same shape, but different energy
10. Which of the following statements are not correct ?
 a) The ionization energy of a hydrogen-like species in its ground state is equal to the magnitude of energy of the orbit having $n = 1$
 b) The ionization energy of a hydrogen-like species in its ground state increases in proportion to the positive charge in its nucleus
 c) According to the uncertainty principle, $\Delta p \Delta x \leq h/4\pi$
 d) The energy of an electron in an orbital of a multielectron atom depends only on the principal quantum number n

11. For the energy levels in an atom, which one of the following statement(s) is (are) correct?
- There are seven principal electron energy levels
 - The second principal energy level can have four sub-energy levels and contain a maximum of eight electrons
 - The M energy level can have a maximum of 32 electrons
 - The 4s sub-energy level is at a lower energy than the 3d sub-energy level.
12. Which of the following statements are correct ?
- The angular momentum of an electron due to its spinning is given as $\sqrt{s(s+1)} \frac{h}{2\pi}$ (where s can take the value of 1/2)
 - The angular momentum of an electron due to its spinning is given as $m_s(h/2\pi)$ where m_s can take the value of $\pm 1/2$.
 - According to Pauli's exclusion principle, no two electrons in an atom can have the same values of quantum numbers n, l and m
 - The azimuthal quantum number cannot have negative values.
13. The chlorine atom has
- 6 electrons with $l = 0$
 - 11 electrons with $l = 1$
 - 5 orbitals with $m = 0$
 - 2 orbitals with $m = +1$
14. Which of the following sets of quantum numbers represents the orbitals of same shape and orientation in external field.
- $n = 3, l = 2, m = +1, s = +1/2$
 - $n = 3, l = 2, m = -1, s = -1/2$
 - $n = 4, l = 2, m = +1, s = -1/2$
 - $n = 3, l = 2, m = 0, s = +1/2$
15. Choose the correct statement(s):
- The shape of an atomic orbital depends upon azimuthal quantum number
 - The orientation of an atomic orbital depends upon the magnetic quantum number
 - The energy of an electron in an atomic orbital of multi-electron atom depends on principal quantum number only
 - The number of degenerate atomic orbitals of one type depends upon the value of azimuthal quantum number
16. Select the correct configurations among the following
- $\text{Cr}(Z = 24): [\text{Ar}]3d^5, 4s^1$
 - $\text{Cu}(Z = 29): [\text{Ar}]3d^{10}, 4s^1$
 - $\text{Pd}(Z = 46): [\text{Kr}]4d^{10}, 5s^0$
 - $\text{Pt}(Z = 78): [\text{Xe}]4d^{10}, 5s^2$

Linked Comprehension Type Questions

Passage :

The substances which contain species with unpaired electrons in their orbitals behave as paramagnetic substances. Such substances are weakly attracted by the magnetic field. The paramagnetism is expressed in terms of magnetic moment. The magnetic moment is related to the number of unpaired electrons according to the following relation : Magnetic moment, $\mu = \sqrt{n(n+2)} \text{ B.M.}$

Where 'n' = number of unpaired electrons. B.M stands for Bohr magneton, a unit of magnetic moment.

17. Which of the following has the highest magnetic moment ?
- Fe^{+2}
 - Mn^{+2}
 - Cr^{+3}
 - V^{+3}

18. An ion of a d-block elements has magnetic moment 5.92 B.M. Select the ion among the following.
 a) Zn^{+2} b) Sc^{+2} c) Mn^{+2} d) Cr^{3+}
19. Which of the following is diamagnetic ?
 a) He^{+2} b) Sc^{+3} c) Mg^{+2} d) O^{-2}

Matrix Matching Type Questions20. **Column-I****Column-II**

A) Violation of Aufbau's rule

P) $\uparrow\downarrow$ $\uparrow\uparrow$ $\uparrow$ $\square$

B) Violation of Pauli's exclusion principle

Q) $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\square$

C) Violation of Hund's rule

R) $\uparrow\downarrow$ $\uparrow\uparrow$ $\uparrow$ $\uparrow$

D) Violation of both Pauli's and Hund's rule

S) $\uparrow$ $\uparrow\downarrow$ $\uparrow$ $\uparrow$ **Integer Type Questions**

21. The number of electrons with $m = 0$ in an atom with atomic number 33 is $5x$. Then the value of x is ____
22. Find out the number of angular nodes in the orbital to which the last electron of Cr enter
23. The orbital angular momentum of 3p electron is $\sqrt{x} \frac{h}{2\pi}$. Then the value of x is ____.
24. An ion Mn^{a+} has the magnetic moment equal to 4.9 BM. What is the value of a ?
25. A compound of vanadium has magnetic moment of 1.73 BM. It suggests the vanadium oxidation state in the compound is $+x$. Then the value of x is ____.

KEY SHEET (LECTURE SHEET)**EXERCISE-I****LEVEL-I**

- 1) 3 2) 1 3) 4 4) 2 5) 4 6) 3 7) 3 8) 3
 9) 4 10) 1 11) 3 12) 4 13) 2 14) 4 15) 4 16) 1
 17) 1 18) 1 19) 4 20) 3 21) 2 22) 2 23) 3 24) 3
 25) 1

LEVEL-II

- 1) b 2) c 3) d 4) d 5) b 6) b 7) d 8) d
 9) a 10) a 11) d 12) b 13) c 14) d 15) d 16) a
 17) b 18) b 19) ad 20) acd 21) bc 22) cd 23) bc 24) acd
 25) a 26) d 27) a 28) d 29) b 30) A-S; B-R; C-Q; D-P
 31) A-Q; B-PRS; C-PRS; D-Q 32) A-PQ; B-Q; C-R; D-S
 33) A-R; B-Q; C-P; D-S 34) 2 35) 5 36) 5 37) 2 38) 2
 39) 3 40) 2

EXERCISE-II

LEVEL-I

- 1) 4 2) 1 3) 2 4) 1 5) 3 6) 4 7) 3 8) 4
9) 4 10) 1 11) 3 12) 2 13) 2 14) 1 15) 4

LEVEL-II

- 1) b 2) a 3) a 4) c 5) a 6) c 7) a 8) c
9) c 10) b 11) b 12) b 13) c 14) c 15) abc 16) ad
17) bc 18) bcd 19) abcd 20) abc 21) bcd 22) bc 23) b 24) b
25) a 26) a 27) A-PQRS; B-QRS; C-R; D-RS
28) A-R; B-S; C-Q; D-P 29) 5 30) 4 31) 3

EXERCISE-III

LEVEL-I

- 1) 2 2) 3 3) 4 4) 3 5) 4 6) 2 7) 3 8) 3
9) 3 10) 2 11) 3 12) 3 13) 2 14) 3 15) 2 16) 2
17) 1 18) 2 19) 4 20) 3 21) 1

LEVEL-II

- 1) d 2) d 3) c 4) c 5) c 6) d 7) c 8) a
9) abc 10) bcd 11) ad 12) ad 13) abcd 14) ac 15) abd 16) abc
17) b 18) c 19) b 20) A-S; B-PR; C-PQ; D-P 21) 3 22) 2
23) 2 24) 3 25) 4